



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with the answer sheet. Failure to do so may result in disqualification.

TEST CODE **01238010**

FORM TP 2017021

JANUARY 2017

**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION
PHYSICS**

Paper 01 – General Proficiency

1 hour 15 minutes

17 JANUARY 2017 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) newton
- (B) metre
- (C) kilogram
- (D) second

Sample Answer

- (A) ☒ (B) ☐ (C) ☐ (D) ☐

The best answer to this item is “metre”, so (B) has been shaded.

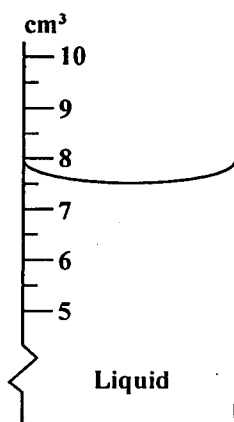
5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
7. You may do any rough work in this booklet.
8. Figures are not necessarily drawn to scale.
9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. How many significant figures are in the quantity 25.92 m^3 ?

(A) 1
(B) 2
(C) 3
(D) 4

Item 2 refers to the following measuring cylinder which is used to determine the volume of a liquid.



2. The volume of the liquid, in cm^3 , is

(A) 7.0
(B) 7.5
(C) 7.8
(D) 8.0

3. In carrying out an experiment to locate a real image using a converging lens, the object can be placed

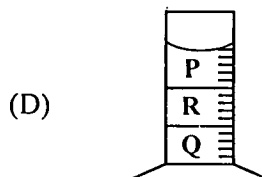
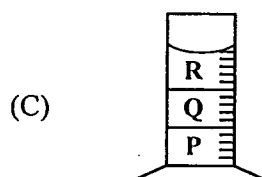
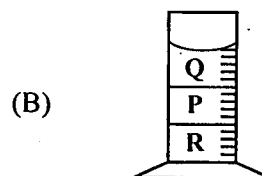
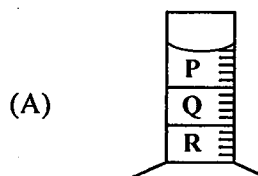
I. between the focal point and the lens
II. at the focal point
III. between the focal point and a point at twice the focal length

(A) I only
(B) II only
(C) III only
(D) I and III only

4. Which of the following instruments is suitable for measuring the diameter of a human hair?

(A) Tape measure
(B) Vernier caliper
(C) Metre rule
(D) Micrometer screw gauge

5. Three immiscible liquids P, Q and R have densities which differ. Q is denser than P but less dense than R. Which of the following diagrams shows how the liquids settle in the measuring cylinder?

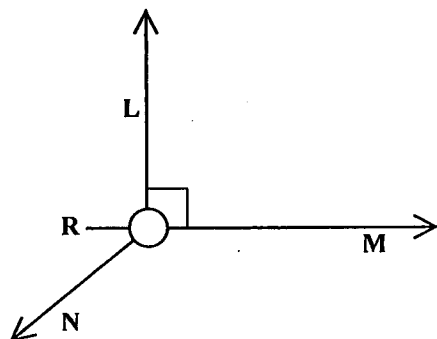


6. The mass of an astronaut is 70 kg when standing on the moon. When he returns to earth his approximate weight will be

(A) 70 kg
(B) 420 kg
(C) 70 N
(D) 700 N

7. When a force F is applied to a spring of original length L , the new length becomes $L + x$. What would be the new length of the spring if a force of $\frac{F}{2}$ was applied instead?
- (A) $L + 2x$
 (B) $2(L + x)$
 (C) $L + x$
 (D) $L + \frac{x}{2}$

Item 8 refers to the following diagram which shows three forces of magnitudes L , M and N , all in the same plane and applied on a ring.



8. Which of the following equations must be TRUE in order for the ring to remain stationary?

- (A) $N^2 = L^2 + M^2$
 (B) $N^2 = L^2 - M^2$
 (C) $L^2 = M^2 + N^2$
 (D) $N = L + M$

9. Linear momentum is

- (A) the product of mass and acceleration
 (B) the product of mass and velocity
 (C) measured in newton-metres
 (D) measured in kilograms per metre

10. Which of the following features must be present in a stable, well-designed racing car?

- (A) Low centre of gravity
 (B) Narrow wheelbase
 (C) Sunroof
 (D) Long front

11. Which of the following statements represent ways in which energy may be used MORE efficiently and economically in homes?

- I. Washing fewer and larger loads in the washing machine
 II. Using a solar water heater instead of an electrical heater
 III. Hanging clothes out to dry in the sun instead of using an electric dryer

- (A) I and II only
 (B) I and III only
 (C) II and III only
 (D) I, II and III

12. When a ball is thrown vertically upwards, and reaches its maximum height, it has

- (A) maximum kinetic energy and maximum potential energy
 (B) maximum kinetic energy and minimum potential energy
 (C) maximum potential energy and minimum kinetic energy
 (D) minimum kinetic energy and minimum potential energy



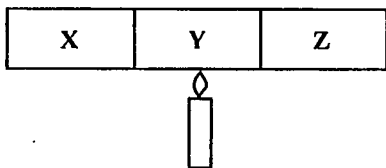
13. A large, heavy parcel is tied with string. It is less painful for a person to pick up the parcel if a cloth is wrapped around the string. This is because the cloth

(A) exerts less force on the fingers
 (B) reduces the tension in the string
 (C) makes the parcel lighter
 (D) reduces the pressure on the fingers

14. The specific latent heat of vaporization of water is $2.26 \times 10^6 \text{ J kg}^{-1}$. When 0.01 kg of water is converted into steam it

(A) absorbs $2.26 \times 10^4 \text{ J}$
 (B) gives out $2.26 \times 10^4 \text{ J}$
 (C) absorbs $2.26 \times 10^8 \text{ J}$
 (D) gives out $2.26 \times 10^8 \text{ J}$

Item 15 refers to the following diagram which shows a copper rod being heated.



15. Which of the following transitions BEST indicates the direction of net energy transfer?

(A) $Y \rightarrow X$ and $Y \rightarrow Z$
 (B) $Y \rightarrow Z$ and $X \rightarrow Z$
 (C) $X \rightarrow Y$ and $Y \rightarrow Z$
 (D) $Z \rightarrow Y$ and $Y \rightarrow X$

16. The heat capacity of a substance is defined as the amount of heat energy

(A) the substance can hold
 (B) 1 kg of the substance can hold
 (C) required to change the substance to another state
 (D) needed to raise the temperature of the substance by 1 degree

17. Under what conditions could the following gas formula be applied to solve problems?

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

I. Mass of gas being constant.
 II. The temperatures T_1 and T_2 must be given in Kelvin.
 III. P_1 and P_2 must be measured in mm of mercury.
 IV. The units of P_1 and V_1 must be the same as those for P_2 and V_2 .

(A) I only
 (B) I and IV only
 (C) I, II and IV only
 (D) I, II, III and IV

18. Which scientist was responsible for arriving at the conclusion that measured amounts of electrical and mechanical energy can be converted to proportionate amounts of heat energy?

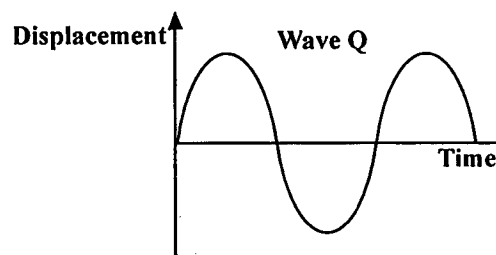
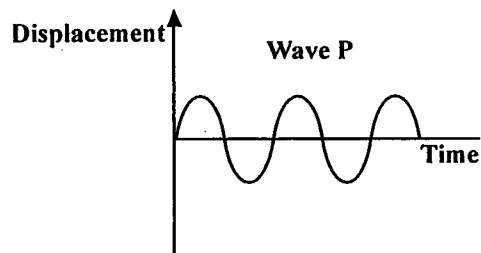
(A) Joule
 (B) Coulomb
 (C) Rumford
 (D) Newton

19. Which of the following methods is the MOST suitable means of heating a brass bob in order to determine its specific heat capacity by the method of mixtures?

(A) Subjecting it to an open flame for ten minutes
 (B) Placing it in contact with an electrical heater for ten minutes
 (C) Placing it in a boiling water bath for ten minutes
 (D) Subjecting it to an infrared beam for ten minutes

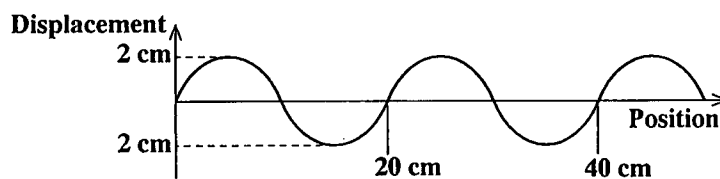
20. A student carries out an experiment to determine the specific heat capacity, c , of a solid using the electrical method. Which of the following equations should be used?
- (A) $c = \frac{M \Delta \theta}{P I}$
- (B) $c = \frac{F \times D}{M \Delta \theta}$
- (C) $c = \frac{IVt}{M \Delta \theta}$
- (D) $c = \frac{M \Delta \theta}{IVt}$
21. The energy required to change the state of a substance was determined to be E_H . If the mass of the substance was DOUBLED, the value of E_H will
- (A) be halved
- (B) be doubled
- (C) be quadrupled
- (D) remain constant
22. The specific latent heat of vaporization of water is the energy required to change 1 kg of water at
- (A) 0°C to ice at 0°C
- (B) 99.9°C to steam at 100.1°C
- (C) 100°C to steam at 100°C
- (D) 0°C to steam at 100°C
23. The 'glasshouse' effect refers to the glasshouse acting as a heat trap. This is a direct result of
- (A) short wavelength infrared rays which are unable to penetrate glass
- (B) long wavelength infrared rays which are radiated from the sun and are unable to penetrate glass
- (C) short wavelength infrared rays which are radiated from the objects within the glasshouse and are unable to penetrate glass
- (D) long wavelength infrared rays which are radiated from the objects within the glasshouse and are unable to penetrate glass
24. The specific latent heat of fusion of water is 340 kJ kg^{-1} . This means that when 10 kg of water freezes
- (A) 34 kJ of heat is absorbed
- (B) 34 kJ of heat is given out
- (C) 3 400 kJ of heat is absorbed
- (D) 3 400 kJ of heat is given out

Item 25 refers to the following graphs (with axes having the same scales) of two sound waves, P and Q.



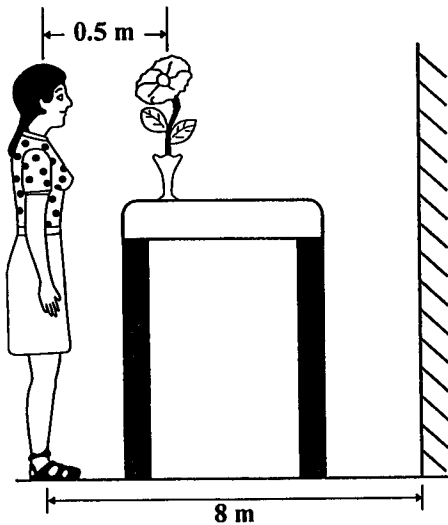
25. Which of the following statements is true?
- (A) P is louder than Q but Q has a higher pitch.
 - (B) P is louder than Q and also has a higher pitch than Q.
 - (C) Q is louder than P but P has a higher pitch.
 - (D) Q is louder than P and also has a higher pitch than P.

Item 26 refers to the following diagram which shows an instantaneous profile of a wave travelling across a water surface.



26. From the information given, the frequency is
- (A) $\frac{1}{20}$ Hz
 - (B) 10 Hz
 - (C) 20 Hz
 - (D) Unknown

Item 27 refers to the following diagram which shows Pam facing a plane mirror which is 8.0 m away from her.



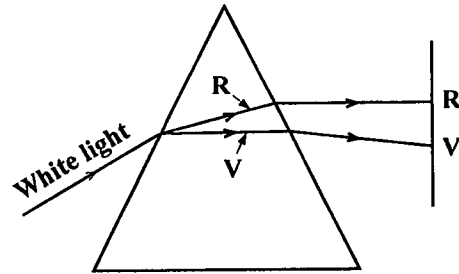
27. Pam views the image of a vase, which is 0.5 m away from her. How far from Pam is the image of the vase?
- (A) 7.5 m
(B) 8.5 m
(C) 15.0 m
(D) 15.5 m

28. An object is placed at the bottom of a tank of water and viewed vertically from above.

Which of the following would be TRUE of the image of the object?

- I. It is virtual.
II. It is diminished.
III. It is nearer to the eye than the object.
- (A) II only
(B) I and II only
(C) I and III only
(D) II and III only

Item 29 refers to the following diagram which shows a beam of white light being dispersed (R - Red, V - Violet) by a prism to form a visible spectrum.

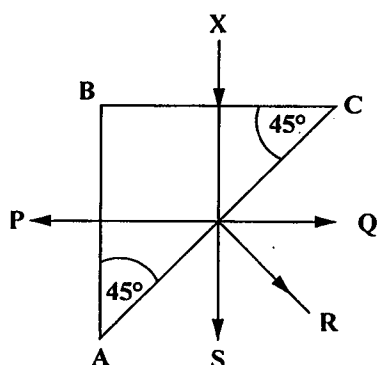


29. Which of the following features make the visible spectrum possible?

- I. The colour violet has the shorter wavelength, hence refracts more than colour red.
II. The colour red has the longer wavelength, hence refracts less than colour violet.
III. The colour violet has the longer wavelength, hence refracts more than colour red.
IV. The colour red has the shorter wavelength, hence refracts more than colour violet.

- (A) I and II only
(B) I and IV only
(C) II and III only
(D) III and IV only

Item 30 refers to the following diagram of a right-angled triangular glass prism, ABC.



30. The critical angle for glass is 42° . X is a ray of light, perpendicular to BC. The ray of light will emerge in the direction of

(A) P
(B) Q
(C) R
(D) S

31. Which of the following waves requires a medium of transmission?

(A) Radio
(B) Infrared
(C) Light
(D) Sound

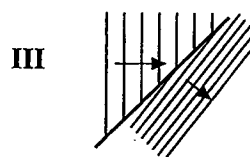
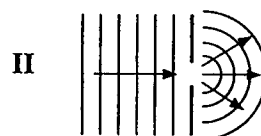
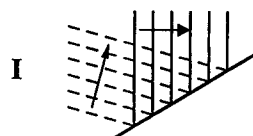
32. The refractive index of a transparent medium with a critical angle, c , for light travelling from the medium to air is

(A) $\frac{1}{c}$
(B) $\frac{90^\circ}{\sin c}$
(C) $\frac{\sin 90^\circ}{\sin c}$
(D) $\sin c$

33. A piece of paper was burnt by adjusting the height of a converging lens above the paper until the most concentrated beam of sunlight was obtained. The distance between the lens and the paper was 10 cm. What was the focal length, in cm, of the lens?

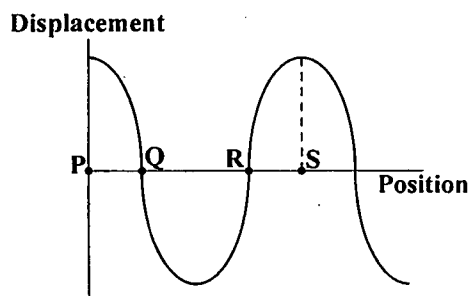
(A) 5
(B) 10
(C) 15
(D) 20

34. Which of the following diagrams could represent diffraction of water waves in a ripple tank?



(A) I only
(B) II only
(C) I and II only
(D) II and III only

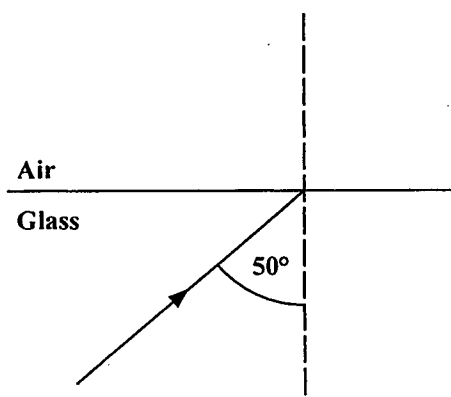
Item 35 refers to the following diagram which shows a transverse wave at a particular instant.



35. The wavelength of the wave is equal to the distance

(A) PQ
(B) PR
(C) PS
(D) QR

Item 36 refers to the following diagram which shows a light ray which is incident at the air-glass interface at an angle of incidence of 50° .



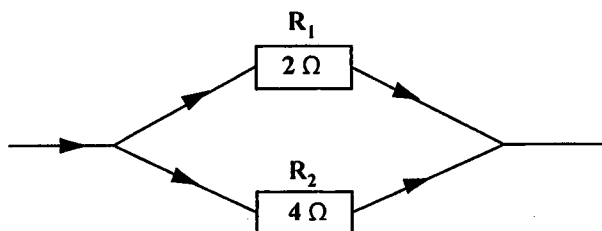
36. If the critical angle for glass is 45° , the light ray will be

(A) reflected
(B) refracted
(C) dispersed
(D) diffracted

37. Which of the following statements is TRUE of a diverging lens?

(A) It is thinner at the centre than at the edge.
(B) It can form only inverted images.
(C) It can form only real images.
(D) It can form only enlarged images.

38. What is the equivalent resistance of the two resistors shown in the following diagram?



(A) 0.5Ω
(B) 1.3Ω
(C) 6Ω
(D) 8Ω

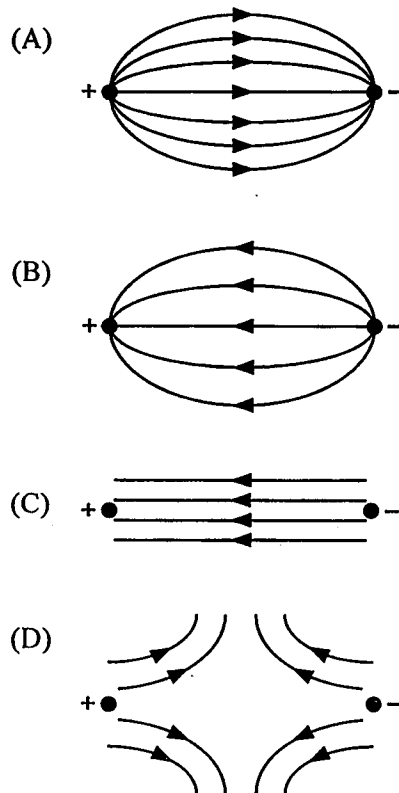
39. A current I flows for a time t between two points at a potential difference of V . The power P expended is equal to

(A) IV/t
(B) IVt
(C) IV
(D) It

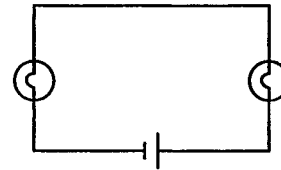
40. A glass rod is rubbed with a piece of silk and is positively charged. The glass rod became charged by

(A) losing protons
(B) losing electrons
(C) gaining protons
(D) gaining electrons

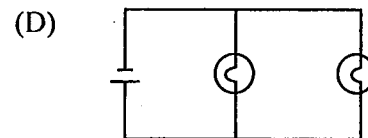
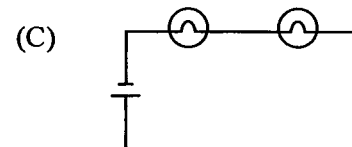
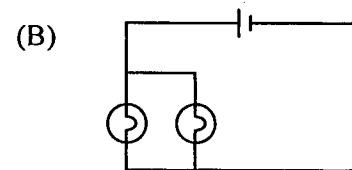
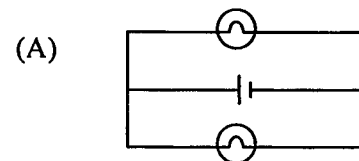
41. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?



Item 42 refers to the following diagram which shows an electrical circuit with a cell and two filament bulbs.



42. Which of the following circuits are electrically the same as the circuit above?



43. The resistance of an IDEAL ammeter is assumed to be

- (A) zero
(B) 0.1 ohm
(C) 1 to 2 ohms
(D) infinite

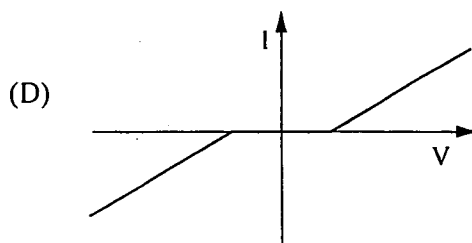
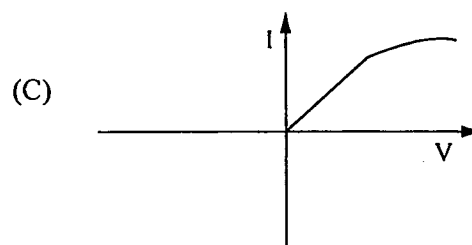
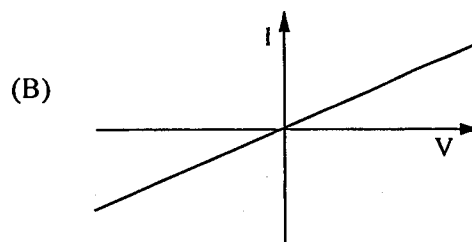
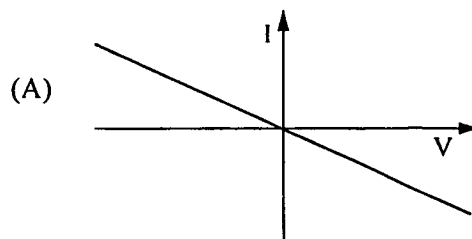
44. Which of the following is NOT one of the ways in which the strength of the magnetic field near a solenoid (long coil) carrying a current can be increased?

- (A) Increasing the resistance of the coil
- (B) Increasing the current in the coil
- (C) Increasing the number of turns per unit length of the coil
- (D) Placing a soft iron core inside the coil

45. Which of the following concerning voltmeters is correct?

	Resistance of a Voltmeter	How it is Connected to the Component
(A)	Low	In parallel
(B)	High	In parallel
(C)	Low	In series
(D)	High	In series

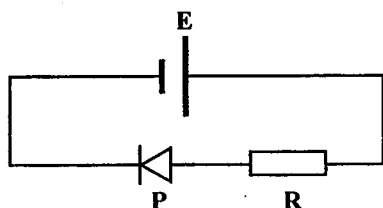
46. Which of the following diagrams represents the current/potential difference relationship for a metallic conductor at a constant temperature?



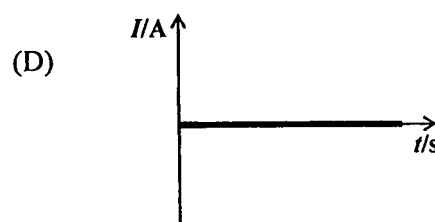
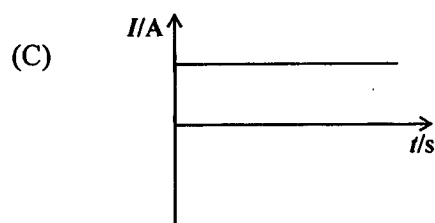
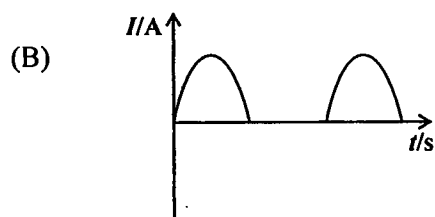
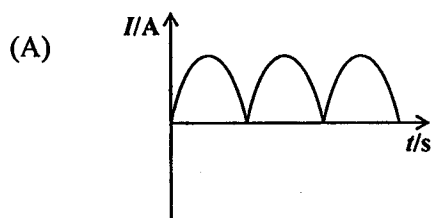
47. An ideal transformer has 1200 turns on the primary coil and 600 turns on the secondary coil. If the voltage across the primary is 20 V, the secondary voltage is

- (A) 2 V
- (B) 5 V
- (C) 10 V
- (D) 20 V

Item 48 refers to the following diagram which shows a cell E, a diode P, and a resistor R, connected in series.



48. Which of the following graphs BEST illustrates the current through R?



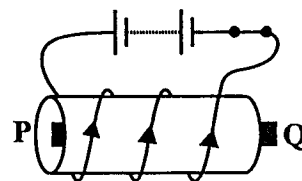
49. Which of the following statements about alternating current is TRUE?

- (A) It can be changed into direct current by a transformer.
- (B) It can be rectified by using a semiconductor diode.
- (C) It can be used to recharge a battery.
- (D) It can be used to transmit electrical energy because of its high frequency.

50. Which of the following statements about insulators is NOT true?

- (A) In an insulator all electrons are bound firmly to their atoms.
- (B) In an insulator many electrons can move freely from atom to atom.
- (C) An insulator can be charged by rubbing.
- (D) A good insulator retains its charge better than a conductor.

Item 51 refers to the following diagram.



51. An iron rod, PQ, is placed inside a plastic pipe wound with a coil as shown in the diagram and a current is passed. When the current is switched off, the rod is likely to have

- (A) positive charge at P, negative charge at Q
- (B) negative charge at P, positive charge at Q
- (C) N pole at P, S pole at Q
- (D) S pole at P, N pole at Q

52. An electromagnet consists of insulated wire wrapped around an iron core.

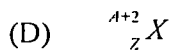
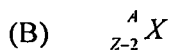
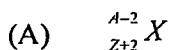
It works because

- (A) iron is always magnetized
- (B) iron is a good electrical conductor
- (C) a magnetic field is produced inside the coil
- (D) an electric field is produced inside the coil

53. The electric power supplied to homes and factories is alternating current (a.c.) rather than direct current (d.c.) because

- (A) the use of a.c. reduces electrical hazards
- (B) the use of a.c. reduces transmission losses
- (C) a.c. enables a wider variety of appliances to be used
- (D) a.c. power lines are less likely to be struck by lightning than d.c. power lines

54. Which of the following would be possible symbols for an isotope of a nuclide represented by A_ZX ?



55. The nuclide ${}^{234}_{90}\text{Th}$ contains

- (A) 90 protons and 234 neutrons
- (B) 235 protons and 90 neutrons
- (C) 90 protons and 144 neutrons
- (D) 144 protons and 90 neutrons

56. Alpha particles, beta particles and gamma rays from a radioactive source are subjected to a transverse magnetic field. Which of them would NOT be deviated from their original path?

- I. Alpha
- II. Beta
- III. Gamma

- (A) II only
- (B) III only
- (C) I and II only
- (D) II and III only

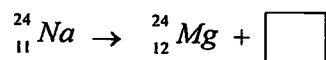
57. A radioactive isotope has a half-life of 20 days. How many days will it take for a given sample to have its activity reduced to $\frac{1}{8}$ of its initial value?

- (A) 1.2 days
- (B) 60 days
- (C) 80 days
- (D) 320 days

58. Which of the following describes two properties of an α -particle?

(A) No charge, very penetrating
 (B) Positive charge, very penetrating
 (C) Negative charge, not very penetrating
 (D) Positive charge, not very penetrating

59. Sodium 24 decays into Magnesium 24 with the emission of a β -particle and can be represented by the following equation.

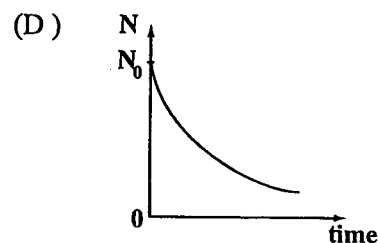
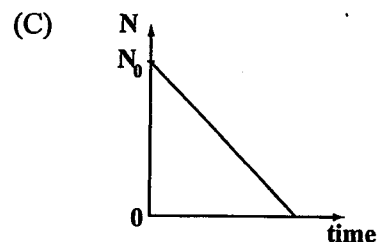
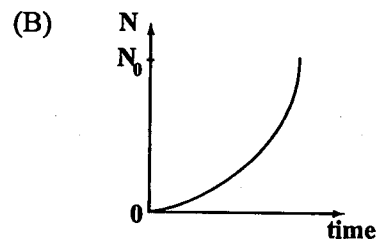
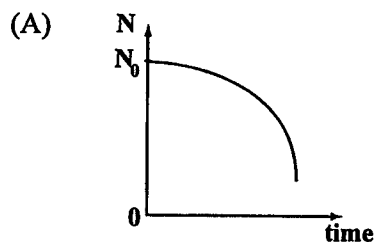


Which of the following should be placed in the box to complete the equation?

(A) ${}_{-1}^0e$
 (B) ${}_{1}^0e$
 (C) ${}_{1}^4\text{He}$
 (D) ${}_{-1}^0\text{He}$

60. No radioactive nuclei are present in a sample at time $t = 0$.

Which of the following graphs BEST represents the variation with time of the number, N , of undecayed nuclei present?



END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.